

BIOLOGY — GRADE 11 | ECOLOGY

Quick Check

Lesson 6: How Many Fish Are Left? Capture-Recapture Simulation to Estimate Population Size

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Ecology | Lesson 6

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Fishers say there are fewer fish in a Kenyan river, but the fish are hidden under the water and keep moving. Why do scientists use a sampling method such as capture-recapture instead of counting every fish?

- A** Fishers are not allowed to count fish
- B** Sampling gives the exact number of fish, but counting does not
- C** Counting every moving fish in a river is not practical, so a sample is used to estimate the total
- D** Fish are too small to be counted by people

In a capture-recapture simulation, 15 counters are captured and marked, then returned and mixed. A second sample of 20 counters is taken and 5 of them are marked. What is the estimated population?

A 60

B 15

C 100

D 75

In the simulation, a group forgets to mix the marked counters thoroughly with the others before taking the second sample. Why is this a problem?

- A** The second sample may not represent the whole population, so the estimate can be badly wrong
- B** The container would hold more than 100 counters
- C** The marked counters would change colour
- D** The formula would no longer have a number for the marked counters

Fisheries officers on Lake Victoria catch and tag 50 Nile tilapia and release them. Later they catch 60 fish, and 12 of them have tags. What is the estimated number of tilapia in the lake?

A 600

B 250

C 500

D 10

Fisheries officers tag 40 fish in a lake. In Year A, they catch 50 fish and 5 are tagged. In Year B, they again catch 50 fish and 10 are tagged. What does this suggest about the fish population in Year B compared with Year A?

- A** It is the same, because the same number of fish were tagged and caught
- B** It cannot be compared, because the estimates are not numbers
- C** It is smaller, because a larger share of the catch is tagged
- D** It is larger, because more tagged fish were found

During a capture-recapture study, many of the tags fall off the tagged fish before the second catch. How will this affect the estimated population size?

- A** There will be no effect, because the formula uses only the second sample
- B** The estimate will be too large, because fewer tagged fish are recognised in the second catch
- C** The estimate will be exact, because tags do not matter
- D** The estimate will be too small, because fewer fish are counted

A class uses capture-recapture on a river and gets an estimate of 500 fish in one year and 250 fish in the next year. Which conclusion is best supported by this evidence?

- A** The estimates suggest the fish population has fallen, but more investigation is needed to find the cause
- B** The estimates prove that overfishing is the cause of the decline
- C** The estimates are exact counts, so exactly 250 fish died
- D** The estimates prove that pollution is the cause of the decline

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).